The easy end never collapsed

graph-colouring reasoning transformer, N = 24 · jobs 49430217 + 49434190

The best model scores EXACTLY 1.000, with 0 failures of 300, at every hint density it was trained on. It only breaks above the training ceiling.

`solve_rate_k0` = 0.07 next to a 0.996 shared-pool rate was logged as a contradiction for two days. It was an arithmetic error in what k = 0 means: that bin asks for 24 hints out of 24, and training never went past 22.

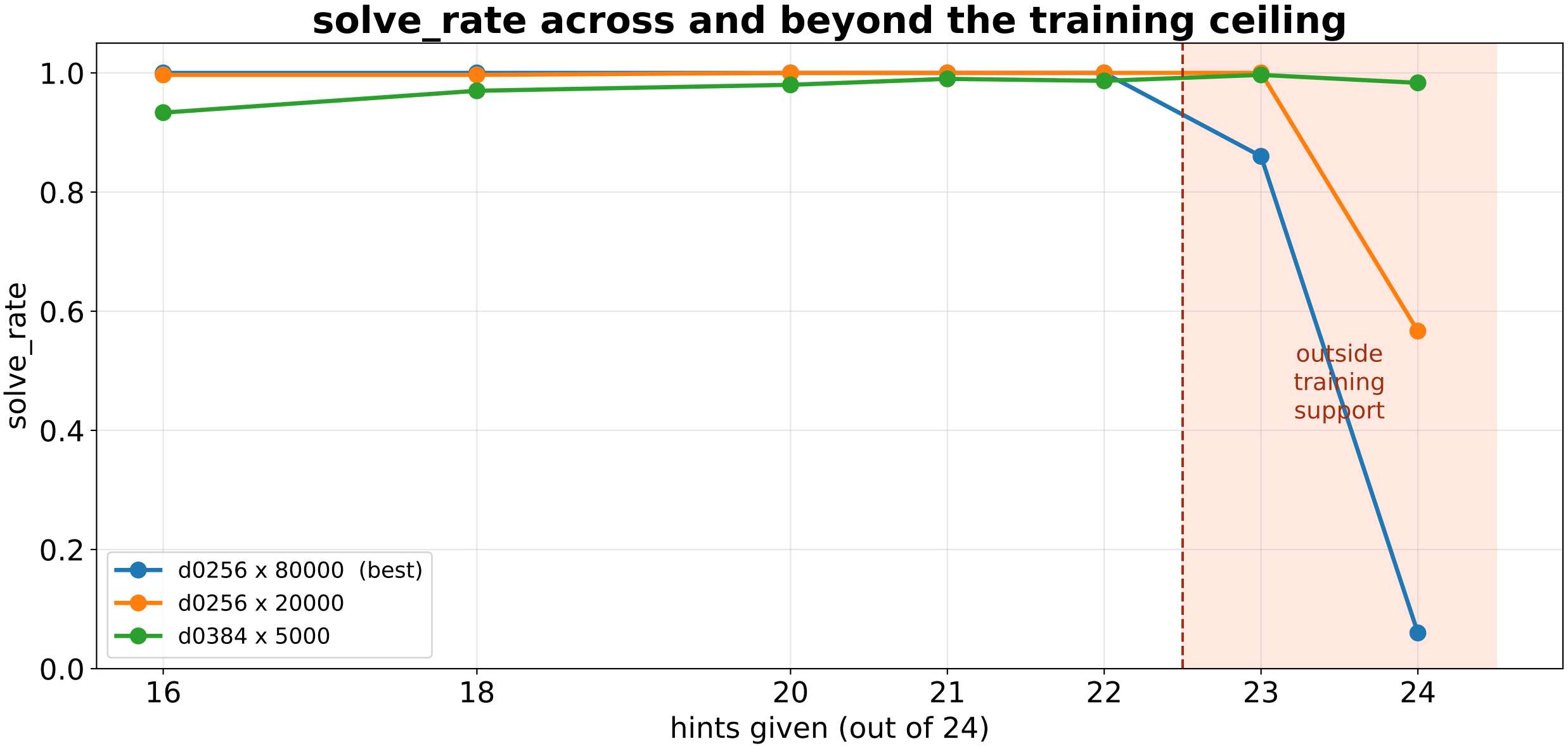
Open problem closed, not a new result: no model was retrained. The fix is to stop reading two out-of-distribution bins as the easy end.

Why k = 0 and k = 1 are out of distribution

k	hints	rho	true DD rounds T	in training support?	solve_rate (best model)
0	24/24	1.0000	0.00	NO	0.060
1	23/24	0.9583	1.00	NO	0.860
2	22/24	0.9167	1.09	yes	1.000
3	21/24	0.8750	1.26	yes	1.000
4	20/24	0.8333	1.52	yes	1.000

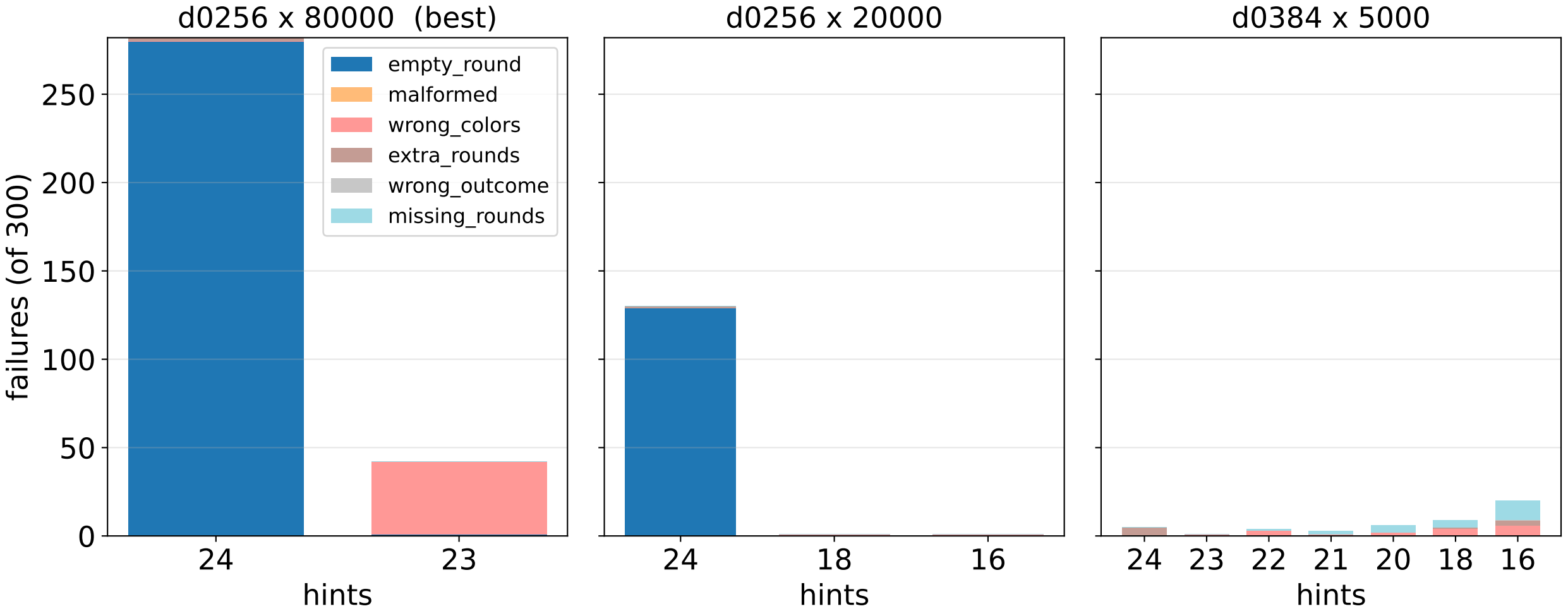
rho_max = 0.95, and removal_path stops at int(floor(rho * N)) = floor(22.8) = 22. mean_o1 averages these five bins, so it mixes two the model has never seen with three it has. k = 0 is doubly strange: with every node hinted the true trace has ZERO deduction rounds, a sequence shape no training instance has.

300 instances per bin, seed_start = 6e10, disjoint from the shared pool (5e10) and from every training seed. Greedy decoding.



The best model (blue) is a flat line at 1.000 across every in-distribution bin and falls only past the dashed line. d0384 x 5000 (green) is the counter-example that keeps this from being an ensemble-wide claim: it handles 24 hints at 0.983 but declines INSIDE distribution, 0.970 at 18 hints and 0.933 at 16.

What the failures actually are



At 24 hints 280 of the best model's 282 failures are `empty_round`: it opens a deduction round on an instance with nothing left to deduce, and the parser raises before the round is appended. That is the predicted out-of-distribution signature. d0384 instead FILLS the round it opens, which is why its 24-hint failures parse and land in `extra_rounds` -- the control showing this is not just relabelling every off-distribution failure. Its in-distribution failures are a third thing again: it emits FEWER rounds than the true trace (2.48 against 2.90 at 16 hints) and colours wrongly, i.e. it truncates the deduction rather than over-running it.

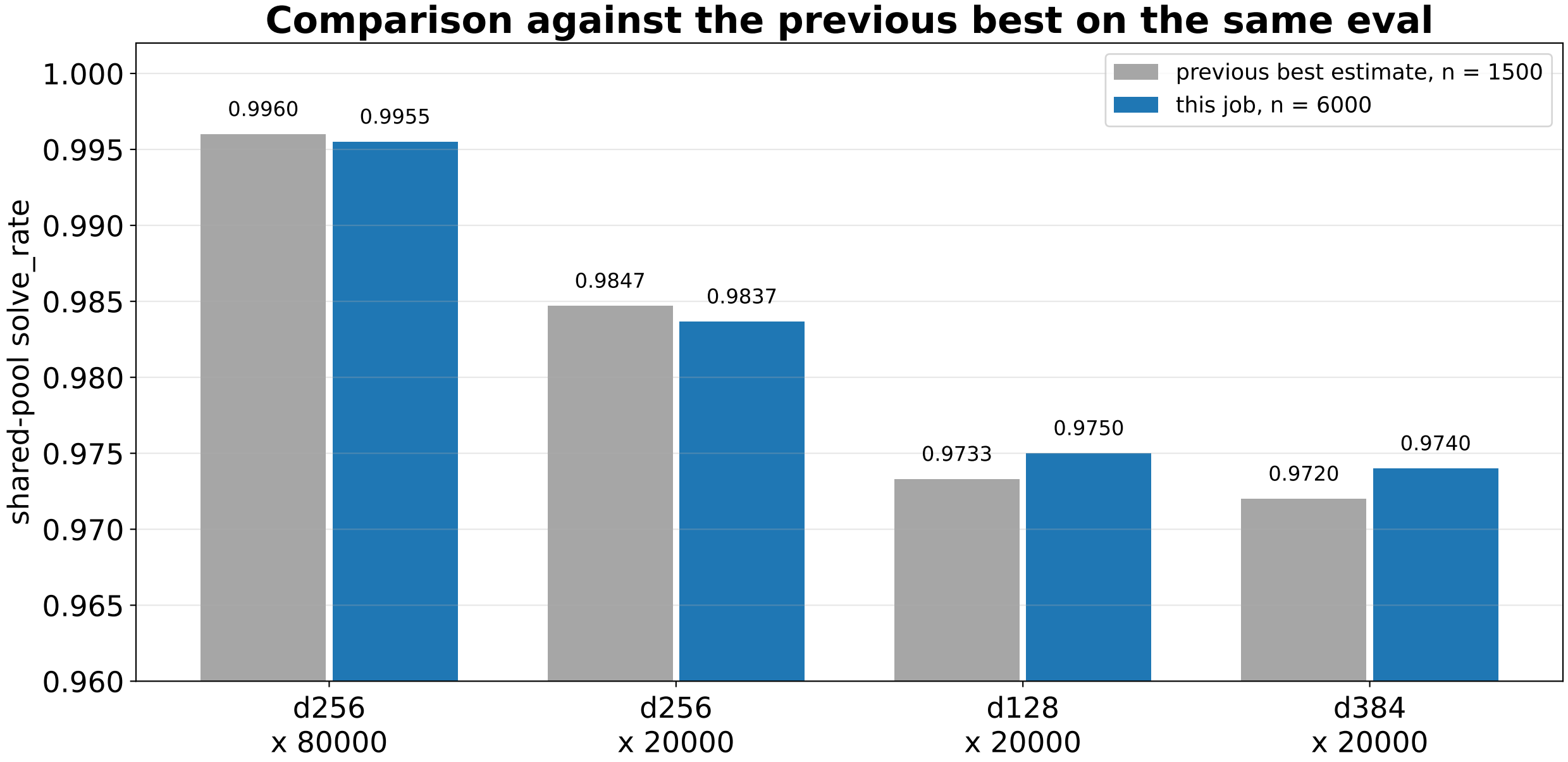
Same job, second question: the shared pool at 4x resolution

model	solve_rate	Wilson 95%	delta vs d256 x 20000	discordant	exact p
d256 x 80000	0.9955	[0.9935, 0.9969]	+0.0118	94 won / 23 lost	2.2e-11
d256 x 20000	0.9837	[0.9801, 0.9866]	reference	--	--
d128 x 20000	0.9750	[0.9707, 0.9787]	-0.0087	76 won / 128 lost	3.3e-04
d384 x 20000	0.9740	[0.9697, 0.9777]	-0.0097	72 won / 130 lost	5.4e-05

n = 6000 is a strict superset of the 1500-instance pool (instance i is seeded at seed_start + i), so this re-scores the same models on the same eval, 4x deeper.

At n_train = 20000, d0128 (4.0M params) and d0384 (16.7M) are still tied with each other (0.9750 vs 0.9740) -- but BOTH are now resolvably below d0256, which n = 1500 could not separate. d_model = 256 is a real optimum at this budget, not a shrug.

McNemar exact, paired on the identical 6000 instances. The reference model's own rate moved 0.9847 -> 0.9837 between pools, i.e. the n=1500 estimate was within its interval.



Every model reproduces its recorded rate to within its Wilson interval, so nothing drifted; what 6000 instances buy is the power to separate the bottom two from d0256. The best model remains d0256 x 80000, now 0.9955 [0.9935, 0.9969].

Scorecard and what this changes

pre-registered	verdict	evidence
(a) in-distribution bins flat and high	HELD	1.000, 0/300 failures, all 5 bins
(b) cliff appears only above 22 hints	HELD	0.860 at 23, 0.060 at 24
(c) >=70% of 24-hint failures are the model deducing where there is nothing to deduce	HELD, after fixing the classifier	280 of 282 (99%) once `empty_round` is split out; it was charged to `malformed` by a test-order bug
(d) the profile is a property of the ensemble, not of one model	MISSED	d0384 x 5000 declines in-distribution (0.933 at 16 hints) while acing 24

Consequences: `mean_o1` and `solve_rate_k0` are retired as competence metrics -- they are extrapolation probes. Nothing needs retraining. The real open question moves to whether network depth must grow with reasoning depth (job 49433454), and to N = 48, where mean DD depth is 6.10 rather than 4.25 (job 49433453).

Checkpoints from jobs 48912548 / 48935109 / 48913017, config d_model x nhead 4 x L12 x ff1024, lr 1e-3, pre-norm, 300 epochs, N=24 d=5 q=4, rho in [0.3, 0.95], hint_mode=removal.